

Stock Market Analysis using Python, MySQL & Power BI

Project Overview

The Stock Market Analysis project is an end-to-end data analytics solution developed to collect, analyze, and visualize historical stock market data. It uses the Alpha Vantage API to fetch daily stock prices, stores the data in MySQL, processes it using Python, and presents interactive insights through a Power BI dashboard.

Problem Statement

Manual analysis of stock market data is difficult due to continuous price fluctuations and large volumes of information. This project automates data collection, storage, analysis, and visualization to support better investment decisions.

Objectives

- Collect stock market data using Alpha Vantage API.
- Store data in MySQL.
- Clean and analyze data using Python.
- Build an interactive Power BI dashboard.
- Compare stock performance and identify trends.

Technologies Used

Python, Pandas, Requests, PyMySQL, MySQL, Alpha Vantage API, Power BI.

Key Insights

- Apple (AAPL) showed the highest trading volume among selected stocks.
- Time-series analysis highlighted price trends and market fluctuations.
- Risk and volatility indicators provided investment insights.
- Interactive dashboards simplified stock comparison.

Conclusion

This project demonstrates the integration of Python, MySQL, and Power BI to create a complete stock market analytics solution. It showcases practical skills in API integration, database management, data analysis, and dashboard development.

Future Scope

Integrate real-time streaming data, implement machine learning for stock price prediction, and deploy the dashboard online using Power BI Service or Streamlit.